AWS ECS — Important Interview Questions & Answers

1.What is Amazon ECS?
Amazon ECS is a fully managed container orchestration service in AWS used to run and manage Docker containers.
It helps us deploy, manage, and scale containerized applications.
Simple ga gurthupettuko:
ECS ante AWS lo Docker containers ni run and manage cheyyadaniki use chese service.
Eng: ECS is a service used to run and manage containers in AWS.

2.What is an ECS Cluster?
An ECS Cluster is a logical group where ECS tasks and services run.
It provides a place to organize and manage our container workloads.
Simple ga gurthupettuko:
Cluster ante ECS tasks and services run ayye logical group.
Eng: An ECS Cluster is a logical group where ECS workloads run.

3.What is an ECS Service?
An ECS Service helps maintain a specified number of tasks continuously.
If a task stops, the service can start a replacement task to maintain the desired count.
Simple ga gurthupettuko:
Service ante required number of ECS tasks ni continuously maintain cheyyadaniki use chestham.
Eng: An ECS Service maintains the required number of running tasks.

4.What is an ECS Task?
An ECS Task is a running instance of a task definition.
It represents the actual container or group of containers running in ECS.
Simple ga gurthupettuko:
Task ante Task Definition prakaram actually run ayye container.
Eng: A Task is a running instance of a task definition.

5.What is an ECS Task Definition?
A Task Definition is a configuration that defines how ECS containers should run.
It includes settings such as container image, CPU, memory, ports, environment variables, and networking.
Simple ga gurthupettuko:
Task Definition ante container ela run avvali ani define chese configuration.
Eng: A Task Definition defines how ECS containers should run.

6.What is the difference between Task and Task Definition?
A Task Definition is the configuration or blueprint, while a Task is the actual running container created from that definition.
Simple ga gurthupettuko:
Task Definition ante blueprint, Task ante aa blueprint prakaram run ayye container.
Eng: Task Definition is the blueprint, while Task is the running container.

7.What is an ECS Container Instance?
A Container Instance is an EC2 instance registered with an ECS cluster and used to run ECS tasks.
It is mainly used when we use the ECS EC2 launch type.
Simple ga gurthupettuko:
Container Instance ante ECS Cluster ki register ayina EC2 instance.
Eng: A Container Instance is an EC2 instance registered with an ECS cluster.

8.What are ECS Capacity Providers?
Capacity Providers help ECS manage the compute capacity used to run tasks.
They can be used with EC2 capacity or AWS Fargate.
Simple ga gurthupettuko:
Capacity Provider ante ECS tasks ki required compute capacity ni manage cheyyadaniki use chestham.
Eng: Capacity Providers help manage the compute capacity used by ECS tasks.

9.What is ECS Fargate?
AWS Fargate is a serverless compute engine for containers.
It allows us to run ECS tasks without managing the underlying EC2 servers.
Simple ga gurthupettuko:
Fargate ante EC2 servers manage cheyyakunda containers run cheyyadaniki use chestham.
Eng: Fargate runs ECS containers without managing EC2 servers.

10.What is the difference between ECS EC2 and Fargate?
With ECS EC2, we manage the underlying EC2 instances.
With Fargate, AWS manages the underlying infrastructure and we mainly manage the containers and task configuration.
Simple ga gurthupettuko:
EC2 launch type lo servers maname manage chestham, Fargate lo AWS infrastructure manage chestundi.
Eng: ECS EC2 requires server management, while Fargate manages the underlying infrastructure.

11.How does ECS Service work?
An ECS Service starts and maintains the required number of tasks based on the desired count.
If a task stops unexpectedly, the service can start a replacement task.
Simple ga gurthupettuko:
Service desired count maintain chestundi, task fail ayite replacement task start chestundi.
Eng: An ECS Service maintains the desired number of tasks and replaces failed tasks.

12.How does ECS work with a Load Balancer?
An ECS Service can be connected to a load balancer through a target group.
The load balancer sends incoming traffic to healthy ECS tasks.
Simple ga gurthupettuko:
Load Balancer traffic ni healthy ECS tasks ki distribute chestundi.
Eng: A Load Balancer distributes traffic to healthy ECS tasks.

13.What is the difference between ECS Cluster and ECS Service?
A Cluster is the logical place where ECS workloads run.
A Service manages the desired number of tasks running inside the cluster.
Simple ga gurthupettuko:
Cluster ante workloads run ayye place, Service ante tasks ni manage chese component.
Eng: A Cluster provides the place for workloads, while a Service manages running tasks.

14.How would you deploy a container using ECS?
First, I would create a task definition with the required container image and settings.
Then I would create an ECS service or run a task inside a cluster.
Simple ga gurthupettuko:
First Task Definition create chesi, tarvata Cluster lo Service or Task run chestham.
Eng: I would create a task definition and then run it using an ECS service or task.

15.What happens if an ECS task fails?
If a task is managed by an ECS Service, ECS can detect the stopped task and start a replacement task based on the desired count.
Simple ga gurthupettuko:
Service lo task fail ayite ECS replacement task ni start chestundi.
Eng: ECS can replace a failed task when it is managed by a service.

16.How would you troubleshoot an ECS task that is not starting?
I would check the task definition, container image, CPU and memory settings, IAM permissions, networking, security groups, and CloudWatch logs.
Simple ga gurthupettuko:
ECS task start avvakapothe Task Definition → Image → CPU/Memory → IAM → Network → Logs check chestham.
Eng: For an ECS task issue, I would check the task definition, image, resources, permissions, networking, and logs.

17.What is the difference between ECS and EC2?
ECS is used to manage and run containers, while EC2 provides virtual servers.
ECS can use EC2 instances as its compute capacity or use Fargate without managing servers.
Simple ga gurthupettuko:
EC2 virtual server provide chestundi, ECS containers ni manage and run chestundi.
Eng: EC2 provides virtual servers, while ECS manages and runs containers.

18.What is the most important thing to remember about ECS?
ECS is used to run and manage containerized applications in AWS.
The important concepts are Clusters, Services, Tasks, Task Definitions, Container Instances, and Capacity Providers.
Simple ga gurthupettuko:
ECS ante AWS lo containers ni run, manage and scale cheyyadaniki use chese service.
Eng: ECS is used to run, manage, and scale containers in AWS.